CoE model $A(\ell)$ vs Migkas+2020 Fig. 6: harmonic amplitudes, tilt done properly (v2)

Step 0. Chris's variables and the model written in them

r = vision-shell radius ≈ 14.3 Gpc (sources on the shell, $d = r$); D = our distance from the CoE; L = density scale length; M = mass enclosed toward the CoE (spreadsheet: M/M_0); (ℓ_0, b_0) = Galactic longitude and tilt of the CoE anti-axis $\hat{D}$; φ = rotation phase in Amp $\sin(\ell \cdot 2\pi/360 + \varphi)$; φ_2 = phase of the 2ℓ transform $\mathrm{Amp}_2 \sin(2\ell \cdot 2\pi/360 + \varphi_2)$; N = overall normalisation. With $\mu = \hat{n} \cdot \hat{D} = \cos b \cos b_0 \cos(\ell - \ell_0) + \sin b \sin b_0$:

$$\frac{A}{A_{\rm all}} = N\Big[1 + \alpha\,\frac{G\,M(<D)}{H_0^2 D^3}\,(3\mu^2 - 1) + \alpha\,\frac{G\,M(<D)}{H_0^2 D^3}\,\frac{r}{D}\Big(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3\Big)\Big]$$

$$M(<D) = 4\pi\rho_0 L^3\big[2 - e^{-D/L}\big((D/L)^2 + 2D/L + 2\big)\big] \qquad \text{(exponential core, } \rho = \rho_0 e^{-R/L}\text{)}$$

Step 1. Define parameters (built from Chris's variables; only these compact ones are used below)

$$\varepsilon \equiv r/D\ (= d/D), \qquad x \equiv D/L, \qquad M(<D) = 4\pi\rho_0 L^3\big[2 - e^{-x}(x^2 + 2x + 2)\big]$$

$$M_{\rm crit}(<D) = \tfrac{4\pi}{3}\rho_c D^3 = \frac{H_0^2 D^3}{2G} \quad (\rho_c = 3H_0^2/8\pi G)$$

$$\kappa \equiv \frac{G\,M(<D)}{H_0^2 D^3} = \frac{M(<D)}{2M_{\rm crit}(<D)} = \frac{3}{2}\,\frac{\rho_0}{\rho_c}\,g(x), \qquad g(x) = \frac{2 - e^{-x}(x^2 + 2x + 2)}{x^3} \quad \big(g \to \tfrac{1}{3} \text{ as } x \to 0\big)$$

$\alpha = O(1)$ time-integration factor, $\qquad K \equiv \alpha\kappa$

$$\frac{A}{A_{\rm all}} = N\big[1 + K(3\mu^2 - 1) + K\varepsilon\big(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3\big)\big] \qquad \text{(unknowns: } N,\ K,\ \varepsilon,\ b_0;\ \ \ell_0 \text{ fixed below)}$$

Step 2. How the tilt enters: average over each Migkas sector (assumptions: 90° longitude sectors centred every 5°, full latitude range minus the $|b| \le 20°$ plane cut of the cluster sample, area weight $\cos b\, db\, d\ell$, clusters treated as uniform on the sky)

Write $a = \cos b_0,\ s_0 = \sin b_0,\ \psi = \ell_c - \ell_0$. The cut is symmetric in b, so odd powers of $\sin b$ average to zero and the averages factorise:

$$\langle\mu\rangle = a\langle c\rangle\langle C\rangle, \qquad \langle\mu^2\rangle = a^2\langle c^2\rangle\langle C^2\rangle + s_0^2\langle s^2\rangle, \qquad \langle\mu^3\rangle = a^3\langle c^3\rangle\langle C^3\rangle + 3as_0^2\langle cs^2\rangle\langle C\rangle$$

($c = \cos b,\ s = \sin b,\ C = \cos(\ell - \ell_0)$). Latitude moments, $20° < |b| < 90°$, weight $\cos b$:
$\langle c\rangle = 0.68417,\ \langle c^2\rangle = 0.51367,\ \langle c^3\rangle = 0.40530,\ \langle s^2\rangle = 0.48633,\ \langle cs^2\rangle = 0.27887.$
Sector (longitude) average: $\langle\cos m(\ell - \ell_0)\rangle = W_m \cos m\psi$, $W_m = \frac{\sin(m\pi/4)}{m\pi/4}$:
$W_1 = 0.90032,\ W_2 = 0.63662,\ W_3 = 0.30011;\quad \langle C^2\rangle = \frac{1}{2} + \frac{1}{2}W_2\cos 2\psi,\ \langle C^3\rangle = \frac{3}{4}W_1\cos\psi + \frac{1}{4}W_3\cos 3\psi$. Result:

$$\Big\langle\frac{A}{N}\Big\rangle(\psi) = 1 + Kq_0 + K\varepsilon\, o_1\cos\psi + Kq_2\cos 2\psi + K\varepsilon\, o_3\cos 3\psi$$

$$q_0 = \tfrac{3}{2}a^2\langle c^2\rangle + 3s_0^2\langle s^2\rangle - 1 = 0.45900 - 0.68850\,a^2, \qquad q_2 = \tfrac{3}{2}a^2\langle c^2\rangle W_2 = 0.49052\,a^2,$$
$$o_1 = aW_1\big(\tfrac{9}{2}\langle c\rangle - \tfrac{45}{8}a^2\langle c^3\rangle - \tfrac{45}{2}s_0^2\langle cs^2\rangle\big) = 0.90032\,a\,(-3.1958 + 3.9948\,a^2), \quad o_3 = -\tfrac{15}{8}a^3\langle c^3\rangle W_3 = -0.22806\,a^3.$$

Checked against a brute-force 2-D sector average (grid $0.25° \times 0.5°$) at $b_0 = 30°$: agreement to $< 10^{-6}$; all $\sin m\psi$ terms vanish.

b_0	q_0	q_2	o_1	o_3	$\lvert o_1/o_3\rvert$	old shortcut o_1 ($\mu = a\cos\psi$, $\times W_1$)
0°	−0.2295	0.4905	+0.7193	−0.2281	3.154	−1.0129
10°	−0.2087	0.4757	+0.6016	−0.2178	2.762	−0.8471
19.10°	−0.1558	0.4380	+0.3159	−0.1924	**1.642**	−0.4448
26.57°	−0.0918	0.3924	0.0000	−0.1632	0	0.0000
31.66°	−0.0399	0.3554	−0.2309	−0.1407	**1.642**	+0.3252
40°	+0.0550	0.2878	−0.5873	−0.1025	5.729	+0.8270

The latitude spread flips the sign of o_1 relative to the old shortcut at every $b_0 \neq 26.57°$ (both vanish at $\cos^2 b_0 = 0.8$).

Step 3. Chris's transforms, written as equations

Each model harmonic is given its own phase; only its amplitude is kept from the physics. Model harmonic m with signed coefficient M_m has natural peak at ℓ_0 (if $M_m > 0$) or $\ell_0 + 180°/m$ (if $M_m < 0$):

$$|M_1|\cos(\ell - \ell_0^{(1)}) \;\to\; |M_1|\cos(\ell - \ell_0^{(1)} - \phi_1) = |M_1|\sin(\ell + \varphi), \quad \varphi = 90° - \ell_0^{(1)} - \phi_1 \qquad \text{(rotation)}$$

$$|M_2|\cos 2(\ell - \ell_0) \;\to\; |M_2|\cos 2(\ell - \ell_0 - \phi_2) = |M_2|\sin(2\ell + \varphi_2), \quad \varphi_2 = 90° - 2(\ell_0 + \phi_2) \qquad (2\ell \text{ transform})$$

The octupole is not given its own phase; its offset from the data is reported as a mismatch δ_3.

Step 4. Data amplitudes and phases ($A_{\rm fit} = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell$; refit to the 71 digitised points reproduces these, $R^2 = 0.9150$)

$m = 0:$ $A_0 = 0.99650$

$m = 1:$ $|A_1| = \sqrt{0.0296^2 + 0.0434^2} = 0.05253,\ \ell_1 = \text{atan2}(0.0434,\, 0.0296) = 55.71°$

$m = 2:$ $|A_2| = \sqrt{0.0636^2 + 0.0366^2} = 0.07338,\ 2\ell_2 = \text{atan2}(0.0366,\, 0.0636) = 29.92° \Rightarrow \ell_2 = 14.96°$

$m = 3:$ $|A_3| = \sqrt{0.0299^2 + 0.0114^2} = 0.03200,\ 3\ell_3 = \text{atan2}(-0.0114,\, -0.0299) = 200.87° \Rightarrow \ell_3 = 66.96°$

Axis: $\ell_0 \equiv \ell_2 = 14.96°$ ($\equiv 194.96°$ for m=2; with $K > 0$, $q_2 > 0$ the model quadrupole peaks at ℓ_0). In Chris's form: $0.05253\sin(\ell + 34.30°)$, $0.07338\sin(2\ell + 60.08°)$. ($\ell_3$ is defined mod 120°.)

Step 5. Set amplitudes equal

$$0.99650 = N(1 + Kq_0), \qquad 0.05253 = NK\varepsilon\,|o_1|, \qquad 0.07338 = NK\,q_2, \qquad 0.03200 = NK\varepsilon\,|o_3|$$

Step 6. Solve (4 equations, 4 unknowns N, K, ε, b_0)

(a) m=1 ÷ m=3 removes $NK\varepsilon$ and leaves b_0 alone ($u \equiv a^2 = \cos^2 b_0$):

$$\frac{0.05253}{0.03200} = 1.6417 = \frac{0.90032\,|-3.1958 + 3.9948u|}{0.22806\,u} \;\Rightarrow\; |-3.1958 + 3.9948u| = 0.41585\,u$$

$$\Rightarrow\; u = \frac{3.1958}{3.9948 \mp 0.41585} = \begin{cases} 0.89296 & b_0 = 19.10° \quad (o_1 > 0) \\ 0.72457 & b_0 = 31.66° \quad (o_1 < 0) \end{cases}$$

Two exact roots ($|o_1/o_3|$ falls from 3.15 at $b_0 = 0$ to 0 at 26.57°, then rises), none else in $0 < b_0 < 90°$.

(b)–(d) Branch A, $b_0 = 19.10°$ ($a = 0.94496$, $a^3 = 0.84381$; $q_0 = -0.15580$, $q_2 = 0.43801$, $o_1 = +0.31592$, $o_3 = -0.19244$):

$$NK = \frac{0.07338}{0.43801} = 0.16753, \qquad \varepsilon = \frac{0.03200}{0.16753 \times 0.19244} = \frac{0.03200}{0.032240} = 0.99257,$$

$$N = 0.99650 + 0.16753 \times 0.15580 = 1.02260, \qquad K = \frac{0.16753}{1.02260} = 0.16383$$

Branch B, $b_0 = 31.66°$ ($a = 0.85122$; $q_0 = -0.03987$, $q_2 = 0.35542$, $o_1 = -0.23092$, $o_3 = -0.14066$): $NK = 0.20646$, $\varepsilon = 1.10188$, $N = 1.00473$, $K = 0.20549$.

Plug-back (branch A): $m0$: $1.02260(1 - 0.16383 \times 0.15580) = 0.99650$; $m1$: $0.16753 \times 0.99257 \times 0.31592 = 0.05253$; $m2$: $0.16753 \times 0.43801 = 0.07338$; $m3$: $0.16753 \times 0.99257 \times 0.19244 = 0.03200$ ✓ (residuals $< 10^{-12}$; same for B).

Step 7. Phases (the transforms)

Branch A: $M_1 > 0$, natural dipole peak 14.96° ($\sin(\ell + 75.04°)$) → data 55.71° ($\sin(\ell + 34.30°)$): $\phi_1 = 55.71 - 14.96 = +40.75°$ (0.711 rad), i.e. φ: 75.04° → 34.30° (1.310 → 0.599 rad).

Branch B: $M_1 < 0$, natural dipole peak 194.96°, $\phi_1 = -139.25°$ (a sign flip plus 40.75°).

2ℓ: $\phi_2 = \ell_2 - \ell_0 = 0°$ by construction; $\varphi_2 = 60.08°$ (1.049 rad), unchanged from the model.
Octupole: $M_3 < 0$ in both branches, natural peak $\ell_0 + 60° = 74.96°$; data 66.96° $\Rightarrow$ $\delta_3 = -8.00°$ in ℓ ($-24.0°$ in 3ℓ phase).

R^2 against the 71 points: no transforms: 0.764 (A), -0.069 (B); rotation + 2ℓ (octupole at natural phase): 0.897 (both); octupole also given its own phase: 0.915 (identical to the Fourier fit, by construction).

Step 8. Back to Chris's variables

	Branch A ($b_0 = 19.10°$)	Branch B ($b_0 = 31.66°$)
$D/r = 1/\varepsilon$	1.0075	0.9075
$D = 14.3\,\mathrm{Gpc} \times D/r$	14.41 Gpc = 47.0 Gly	12.98 Gpc = 42.3 Gly
$K = \alpha\kappa$; $M(<D)/M_{\rm crit} = 2K/\alpha$	0.1638; $0.328/\alpha$	0.2055; $0.411/\alpha$
N	1.0226	1.0047
ℓ_0	14.96°	14.96°
rotation ϕ_1	$+40.75° = 0.711$ rad	$-139.25° = -2.430$ rad
Chris's φ (model $\to$ data)	$75.04° \to 34.30°$ (1.310 $\to$ 0.599 rad)	$-104.96° \to 34.30°$
2ℓ phase ϕ_2; φ_2	0°; 60.08° (1.049 rad)	0°; 60.08°
octupole mismatch δ_3	$-8.0°$	$-8.0°$

D/L is not determined: only $\kappa = \frac{3}{2}(\rho_0/\rho_c)g(x)$ enters, so x is degenerate with ρ_0. Branch A, $\alpha = 1$: $\rho_0/\rho_c = 2K/(3g(x)) = 0.353,\ 0.474,\ 0.680,\ 1.351,\ 2.556$ for $x = D/L = 0.1,\ 0.5,\ 1,\ 2,\ 3$.

Caveats. An exact solution exists (two branches) because there are four amplitude equations and four unknowns; matching is automatic and is not a test. ℓ_0, the rotation ϕ_1 and the 2ℓ phase are extra free parameters not predicted by the axisymmetric physics (which has only $\cos m\psi$ terms, one axis); the 41° rotation is the size of the model's failure to put the dipole on the quadrupole axis. The 71 points come from overlapping 90° sectors stepped by 5°, so there are only ∼16 independent values; R^2 overstates the evidence. The analytic tidal form is a small-r/D expansion, but both roots have $r/D \approx 1$ (branch B has $D < r$, $r/D = 1.10$), where it is not reliable. Sector weighting assumes clusters uniform on the sky outside $|b| \le 20°$; the real cluster distribution will shift the q, o coefficients.